



Development of an Empirical Bayesian Single-Attribute Acceptance Sampling Plan for Food Quality using a Beta–Binomial Model

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ABSTRACT

Background: The variation in the quality characteristics of a manufactured product across different lots may arise from variation within lots (batches) or from variation between lots. Within-lot variation is addressed through statistical process control, whereas between-lot variation, which may arise from shifts in the process center, is more appropriately evaluated through acceptance sampling. Traditional acceptance sampling plans in food quality assurance rely primarily on frequentist frameworks that evaluate lots in isolation, thereby ignoring valuable historical process information and failing to adequately address uncertainty in the process center. In food manufacturing, where inspection is often destructive and costly, incorporating historical process information into acceptance decisions is essential. Bayesian methods provide an effective framework for integrating prior knowledge with current sample data.

Objective: This study develops and evaluates a Bayesian single-attribute acceptance sampling plan using a Beta prior distribution to model the lot proportion nonconforming, p , and benchmarks its performance against classical frequentist baselines under realistic food safety constraints.

Methods: The number of nonconforming items, Y , in a sample of size n is assumed to follow a Binomial distribution, $Y \sim \text{Binomial}(n, p)$, with a conjugate Beta prior, $p \sim \text{Beta}(\alpha, \beta)$. Method of Moments estimation was applied to simulated continuous production records comprising $K = 100$ lots to calibrate an empirical baseline prior, $\text{Beta}(2, 198)$. The proposed empirical plan was systematically benchmarked against three noninformative priors: Haldane, $\text{Beta}(0.001, 0.001)$; Jeffreys, $\text{Beta}(0.5, 0.5)$; and Uniform, $\text{Beta}(1, 1)$, as well as three informative regimes: Low Compliance, $\text{Beta}(2, 38)$; Conservative, $\text{Beta}(5, 95)$; and Optimistic, $\text{Beta}(1, 199)$. Plan performance was evaluated using Operating Characteristic (OC) curves, Bayes risk, Average Total Inspection (ATI), and prior sensitivity analysis.

Results: The empirical $\text{Beta}(2, 198)$ prior reduced the required sample size by 33.3% while maintaining Producer's Risk ($\alpha_0 \leq 0.05$ at $\text{AQL} = 1\%$) and Consumer's Risk ($\beta_0 \leq 0.10$ at $\text{LQL} = 5.0\%$) within the specified boundaries. The OC curve of the empirical Bayesian plan closely matched the discrimination profile of the classical frequentist baseline while requiring substantially less inspection effort. Sensitivity analysis confirmed the robustness of the proposed plan to perturbations in the prior hyperparameters, while dynamic Bayesian updating provides a mechanism for maintaining consumer protection when process conditions change.

Conclusion: The conjugate Beta–Binomial Bayesian framework improves sampling efficiency and decision-making by combining historical process stability with current inspection data. The proposed adaptive approach provides a mathematically sound framework for food quality assurance, potentially reducing laboratory inspection costs while maintaining specified food safety requirements.

Key words: Bayesian acceptance sampling, Beta prior distribution, Beta-binomial model, Food quality assurance, Operating characteristic curve, Bayes risk

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INTRODUCTION

In manufacturing industries, raw materials are typically taken and processed in production lots. Product quality can vary due to differences in raw materials, production processes, machinery, operating conditions, handling systems, and environmental factors. Measured quality characteristics may vary due to differences within a lot or between lots. While within-lot variation is typically monitored and controlled using Statistical Process Control (SPC), between-lot variation affects the overall quality of submitted batches and must be evaluated through acceptance sampling. Particularly in food production, 100% inspection is rarely feasible because destructive testing

and laboratory analysis are often costly or impractical. Thus, acceptance sampling provides an effective statistical method for deciding whether to accept or reject a production lot based on sample inspection.

Classical acceptance sampling plans, such as MIL-STD-105E (Department of Defense, 1989) and ISO 2859-1 (International Organization for Standardization, 1999), treat the proportion of nonconforming items in a lot, p , as a fixed but unknown parameter. These plans evaluate submitted lots using predefined sampling schemes based on specified quality levels, making decisions from the observed number of nonconforming items in the current sample. They are designed to satisfy predetermined re-

quirements for Producer's Risk (α), Consumer's Risk (β), Acceptance Quality Level (AQL), and Limiting Quality Level (LQL). Although these methods have been widely adopted in industrial quality assurance, they treat each lot in isolation and do not incorporate historical information regarding process performance and supplier reliability into the sampling decision.

In practice, variations in quality arise not only from sampling fluctuations but also from changes in the production process. The quality of lots produced at different times often varies because process centers can shift due to factors such as raw material batches, machinery wear and tear, changes in operators, or environmental conditions. Consequently, variation between lots may be greater than variation within lots (Facchinetti et al., 2022). Treating all submitted lots as independent observations from a fixed process overlooks the inherent uncertainty associated with process variation. This may result in inefficient sampling plans, including unnecessarily large sample sizes and suboptimal decisions regarding lot acceptance (Facchinetti et al., 2022; Vijayaraghavan and Sakthivel, 2010).

Bayesian statistics provides a coherent probabilistic framework for dealing with uncertainty in process quality. It combines prior information with observed data to update knowledge and support quality assessment and decision-making. The basic idea underlying the Bayesian method is that uncertainty about an unknown quantity of interest is represented probabilistically through plausible values of that quantity (Khatiwada and Sthapit, 2008). In acceptance sampling, treating the lot proportion nonconforming, p , as a random variable rather than a fixed parameter allows the Bayesian approach to represent pre-existing process knowledge through a prior probability distribution. This prior distribution is updated using sample inspection data through Bayes' theorem, yielding a posterior distribution that reflects the combined information. In single-attribute acceptance sampling, where the number of nonconforming items follows a Binomial distribution, the Beta distribution serves as the natural conjugate prior. The resulting Beta-Binomial model enables the combined use of historical production records and current inspection outcomes, thereby improving sampling plan parameters and reducing unnecessary inspection effort.

The application of Bayesian acceptance sampling is particularly important in food manufacturing, where quality inspection often involves destructive testing, microbiological examination, chemical analysis, and significant laboratory costs. Reliable decisions regarding food safety requirements must therefore be made while minimizing unnecessary inspection efforts. When historical production records demonstrate consistent manufacturing performance, incorporating prior process knowledge can significantly reduce inspection costs without compromising consumer protection (Khatiwada, 2009).

The application of Bayesian methods in quality control and acceptance sampling has grown considerably over time. Early contributions by Hald (Hald, 1960, 1964, 1968, 1977, 1981), Wetherill and Chiu (Wether-

ill and Chiu, 1974), Pande (Pande, 1974), Maxim et al. (Maxim et al., 1976), Calabrese (Calabrese, 1995), Wolfinger (Wolfinger, 1998), Chen and Lam (Chen and Lam, 1999), Dumićić et al. (Dumićić et al., 2006), and Colosimo and del Castillo (Colosimo and del Castillo, 2007) laid important theoretical foundations for Bayesian approaches to quality control and acceptance sampling. More recent studies (Ciccirelli et al., 2025, 2026; Dewi et al., 2022; Facchinetti et al., 2022; Hafeez and Aziz, 2022; Hafeez et al., 2023, 2025; Kadir and Rahi, 2023) have further demonstrated the potential benefits of integrating historical process information into acceptance decisions through Bayesian methods.

Despite these developments, relatively few studies have systematically examined the impact of alternative Beta prior specifications on the performance of Bayesian acceptance sampling plans designed for food quality assurance. In particular, the comparative performance of data-driven empirical priors against noninformative baselines and informative operational regimes under realistic production structures has not been fully assessed. This represents an important methodological gap because the choice of prior distribution can substantially influence posterior inference, sampling requirements, acceptance decisions, and the associated inspection costs.

To address this gap, this study develops and evaluates an empirical Bayesian single-attribute acceptance sampling framework for food quality control using the conjugate Beta-Binomial model. Binomial sample data are generated from simulated continuous production records comprising $K = 100$ lots to represent inspection outcomes and to calibrate an empirical Beta(2, 198) prior using the Method of Moments. To systematically investigate the impact of prior choice, the proposed empirical prior is evaluated alongside a range of noninformative baselines, including the Haldane prior, Beta(0.001, 0.001), the Jeffreys prior, Beta(0.5, 0.5), and the Uniform prior, Beta(1, 1), as well as informative operational regimes, including Low Compliance, Beta(2, 38); Conservative, Beta(5, 95); and Optimistic, Beta(1, 199). All plans are rigorously evaluated against classical frequentist baselines using multiple performance measures, including optimal sample size (n), acceptance threshold (c), Operating Characteristic (OC) curves, Average Total Inspection (ATI), Bayes risk under comprehensive cost models, and prior sensitivity analysis. Furthermore, the proposed framework suggests the potential for automated integration with Laboratory Information Management Systems (LIMS), enabling historical laboratory and production information to be incorporated into future acceptance-sampling decisions.

METHODOLOGY

Model Formulation and Likelihood

Let a submitted lot of food products contain a large batch size N , from which a random sample of n items is selected for quality testing. Let Y represent the number of nonconforming (defective) items observed in the sample. Assuming $N \gg n$, the random variable Y given nonconforming p follows a Binomial distribution:

$$P(Y = y | p) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, 2, \dots, n \quad (1)$$

Specification of the Beta Prior Distribution

To model prior knowledge regarding lot quality p , the Beta distribution with shape parameters $\alpha > 0$ and $\beta > 0$ is specified as the prior probability density function:

$$\pi(p; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}, \quad 0 \leq p \leq 1. \quad (2)$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function. The expected prior proportion nonconforming $E[p]$ and prior variance $\text{Var}(p)$ are expressed as:

$$E[p] = \frac{\alpha}{\alpha + \beta}. \quad (3)$$

$$\text{Var}(p) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (4)$$

Empirical Parameter Estimation via Method of Moments

Historical production lots can be treated as independent realizations of a Beta-distributed process parameter, p_i , where α and β characterize the shape of the underlying proportion nonconforming across lots. For each historical lot i , the observed fraction non-conforming is given by

$$\hat{p}_i = \frac{y_i}{n_i}, \quad i = 1, 2, \dots, K, \quad (5)$$

where y_i denotes the number of nonconforming items observed and n_i the number of items inspected in lot i .

Given historical lot inspection records across K previous lots with observed sample proportions, the hyperparameters α and β of the Beta distribution are estimated using the method of moments, where the theoretical mean and the variance of the Beta distribution are equal to the corresponding sample mean $\bar{p}$ and sample variance s^2 . This yields the following closed-form estimators (Rahman and Amin, 2024):

$$\hat{\alpha} = \bar{p} \left[\frac{\bar{p}(1-\bar{p})}{s^2} - 1 \right]. \quad (6)$$

$$\hat{\beta} = (1-\bar{p}) \left[\frac{\bar{p}(1-\bar{p})}{s^2} - 1 \right]. \quad (7)$$

where

$$\bar{p} = \frac{1}{K} \sum_{i=1}^K \hat{p}_i \quad \text{and} \quad s^2 = \frac{1}{K-1} \sum_{i=1}^K (\hat{p}_i - \bar{p})^2.$$

When $Sp^2 \geq \bar{p}(1-\bar{p})$, the Method of Moments estimator for Beta parameters can fail (yielding negative values). The historical lot variation in this study satisfies $Sp^2 < \bar{p}(1-\bar{p})$.

The resulting estimates $\hat{\alpha}$ and $\hat{\beta}$ constitute an empirically derived, data-driven prior that reflects the historical process behaviour across lots, and were subsequently used to update the posterior distribution of the lot proportion nonconforming upon incorporation of current sample data.

Posterior Distribution and Predictive Probability

Posterior Distribution

As sample inspection data yields y non-conforming items out of n inspected units, Bayes' Theorem combines the Binomial likelihood $P(Y = y | p)$ with the Beta prior density to derive the updated posterior distribution:

$$\pi(p | y) = \frac{P(y | p)\pi(p)}{\int_0^1 P(y | p)\pi(p) dp}. \quad (8)$$

Substituting the Binomial probability mass function and the Beta probability density function gives

$$\pi(p | y) \propto \binom{n}{y} p^y (1-p)^{n-y} \frac{p^{\alpha-1} (1-p)^{\beta-1}}{B(\alpha, \beta)}. \quad (9)$$

By dropping terms that do not depend on p ,

$$\pi(p | y) \propto p^{\alpha+y-1} (1-p)^{\beta+n-y-1}. \quad (10)$$

This kernel matches the functional form of a Beta distribution, which is the posterior distribution of the non-conforming proportion p given y , $(p | y)$.

$$p | y \sim \text{Beta}(\alpha', \beta'). \quad (11)$$

where, $\alpha' = \alpha + y$, $\beta' = \beta + n - y$.

The expected posterior proportion nonconforming (posterior mean) is

$$E[p | y] = \frac{\alpha'}{\alpha' + \beta'}. \quad (12)$$

The posterior mode is

$$\text{Mode}(p | y) = \frac{\alpha' - 1}{\alpha' + \beta' - 2}, \quad \alpha' > 1, \beta' > 1. \quad (13)$$

It is not easy to calculate the posterior median for $\text{Beta}(\alpha', \beta')$ using a simple algebraic equation. It is calculated using the inverse Beta CDF (the quantile function) evaluated at $p = 0.5$. In this work it is calculated using the `qbeta` function in R software.

The posterior standard deviation is

$$SD(\sigma_p) = \sqrt{\frac{\alpha'\beta'}{(\alpha' + \beta')^2(\alpha' + \beta' + 1)}}. \quad (14)$$

The 95% Credible Interval is the interval between the 2.5th and 97.5th percentiles of $\text{Beta}(\alpha', \beta')$.

Predictive Distribution

To determine whether a lot is accepted, we calculate the predictive probability of observing y nonconforming items in a future sample of size n , integrating over all possible values of p :

$$P(Y = y) = \int_0^1 P(Y = y | p)\pi(p) dp. \quad (15)$$

i.e.,

$$P(Y = y) = \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (16)$$

It shows that the marginal predictive distribution of nonconforming Y follows a Beta-Binomial distribution.

Under a single-attribute sampling plan (n, c) , a submitted lot is accepted if the observed number of nonconforming items y does not exceed the acceptance threshold c . In other words, n is the required sample size and c is the maximum allowable number of nonconforming items. The overall probability of lot acceptance (P_a) is the cumulative Beta-Binomial probability. The probability of lot acceptance P_a under an acceptance criterion ($y \leq c$) is given by:

$$P_a = P(Y \leq c) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (17)$$

Acceptance Sampling Plan Optimization Parameters

To establish a single-attribute Bayesian acceptance sampling plan, the plan parameters (n, c) must satisfy two fundamental risk constraints defined at specific quality levels:

1. **Producer's Risk Constraint:** The probability of accepting a good lot at the Acceptable Quality Level (AQL = p_1) must be at least $1 - \alpha_0$.
2. **Consumer's Risk Constraint:** The probability of accepting a bad lot at the Limiting Quality Level (LQL = p_2) must not exceed β_0 .

Mathematically, using the Beta-Binomial predictive acceptance probability P_a , these constraints are expressed as:

$$P_a(p_1) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)} \geq 1 - \alpha_0 \quad (18)$$

$$P_a(p_2) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)} \leq \beta_0 \quad (19)$$

Average Total Inspection

For a single sampling plan with sample size n , acceptance number c , and lot size N , the Average Total Inspection (ATI) function incorporates both the sample size n and the full inspection of rejected lots (assuming rectifying inspection, where rejected lots are 100% inspected and non-conforming items replaced):

$$ATI(p) = n + (N - n)[1 - P_a(p)]. \quad (20)$$

At ideal lot quality ($p \rightarrow 0$), $P_a(p) \rightarrow 1$ and $ATI(0) = n$ (minimum inspection load)

At poor lot quality ($p \rightarrow 1$), $P_a(p) \rightarrow 0$ and $ATI(1) = N$ (full lot screening)

In practice, many (n, c) pairs can satisfy the Consumer's Risk and Producer's Risk constraints. For optimization procedure, the unique optimal plan (n^*, c^*)

is selected by minimizing the Average Total Inspection (ATI). The optimization problem is solved numerically by finding the minimum integer sample size n^* and acceptance threshold c^* that satisfy:

$$\min_{n \in \mathbb{N}} ATI(E[p]) \quad \text{subject to} \quad \begin{cases} P_a(p_1) \geq 1 - \alpha_0, \\ P_a(p_2) \leq \beta_0. \end{cases} \quad (21)$$

Where $E[p] = \frac{\alpha}{\alpha + \beta}$ is the expected prior process nonconforming rate derived from the Beta(α, β).

Simulation Scheme and Data Generation Procedures

To evaluate the performance of the proposed Bayesian acceptance sampling plan under controlled operational conditions, a Monte Carlo simulation framework was implemented in R (v4.3.2). Binomial inspection data were generated to simulate continuous lot-by-lot food quality verification. The simulation procedure followed a four-step protocol:

1. **Lot Nonconforming Rate Generation:** A sequence of $K = 100$ historical production lots was simulated, where the true underlying lot nonconforming proportion p_k for batch k ($k = 1, 2, \dots, K$) was drawn from a true process capability distribution. The true process capability distribution was parameterized with mean $\mu_p = 0.010$ and standard deviation $\sigma_p = 0.007$. This choice reflects an in-control manufacturing environment targeted at an Acceptable Quality Level (AQL = 1.0%).
2. **Sampling and Nonconforming Observation:** From each simulated lot k , a random sample of size $n_k = 100$ items were drawn, and the number of nonconforming items y_k was generated via a Binomial process: $y_k \sim \text{Binomial}(n_k, p_k)$.
3. **Empirical Prior Calibration:** The sample mean $\bar{p}$ and sample variance s_p^2 across the K historical inspection records were computed. Using Method of Moments estimation, the empirical Beta prior hyperparameters $(\hat{\alpha}, \hat{\beta})$ were fitted as given in Equations (6) and (7).
4. **Plan Optimization:** Under each prior regime, optimal sampling plan parameters (n, c) were derived by solving the dual risk constraints:

$$P_a(p_1) \geq 0.95 \quad \text{at AQL} = 0.01 \quad (22)$$

$$P_a(p_2) \leq 0.10 \quad \text{at LQL} = 0.05 \quad (23)$$

Operating Characteristic Curve for Performance Evaluation

The probability of lot acceptance $P_a(p)$ for each plan is calculated using its corresponding predictive density function.

For the frequentist plan:

$$P_a^F(p) = \sum_{y=0}^c \binom{n}{y} p^y (1-p)^{n-y}. \quad (24)$$

For Bayesian plans:

$$P_a^B(p) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (25)$$

Bayes Risk Formulation

To obtain Bayes Risk, first the conditional expected loss $L(n, c, p)$ for a lot of size N and quality p is obtained. The loss function contains four primary cost components: sampling cost (C_1), sample defect (nonconforming) rework cost (C_2), screening rework cost (C_3), and external failure cost (C_4) (Montgomery, 2020).

Sampling Cost

Sampling cost C_1 is the cost of inspection of n units. if c_s is the unit inspection cost $C_1 = c_s \cdot n$.

Sample Defect Rework Cost

Sample defect (nonconforming) rework cost C_2 is the cost of rectifying nonconforming items in the sample. If the internal rework cost is C_i , then $C_2 = C_i \cdot n \cdot p$.

Screening Rework Cost

Screening rework cost C_3 is the rework cost for nonconforming items discovered during 100% screening of rejected lots. Thus, $C_3 = C_i \cdot p \cdot (N - n) [1 - P_a(p)]$.

External Customer Failure Cost

External Customer Failure Cost C_4 is the penalty cost associated with undetected nonconforming items in accepted lots. $C_4 = C_a \cdot (N - n) p P_a(p)$, where C_a represents the field replacement or warranty cost ($C_a \gg C_i$).

The sum of these conditional expected losses yields the conditional Bayes Risk function:

$$R(p; n, c) = c_s n + c_i p n + c_i p (N - n) [1 - P_a(p)] + c_a p (N - n) P_a(p) \quad (26)$$

The overall expected Bayes Risk, $B(n, c)$, is obtained by integrating the conditional loss function over the process proportion nonconforming prior distribution:

$$B(n, c) = \int_0^1 R(p; n, c) \pi(p) dp. \quad (27)$$

The optimal Bayesian sampling plan (n^*, c^*) is determined by finding the integer pair that minimizes the total expected risk:

$$(n^*, c^*) = \arg \min_{(n, c)} B(n, c). \quad (28)$$

Prior Sensitivity Analysis and Robustness

To ensure the decision rules remain robust against parameter mis-specification or historical process shifts, a prior sensitivity analysis framework is built. The sensitivity analysis evaluates the impact of mis-specifying prior hyperparameters (α, β) because hyperparameter estimation relies on historical quality records. Minor sample bias or process drift could lead to mis-specified Beta parameters. The robustness evaluation procedure follows four formal steps:

1. **Baseline Parameterization:** Establish the baseline prior distribution $\pi_0(p) \sim \text{Beta}(\alpha_0, \beta_0)$ and its corresponding optimal sampling plan (n_0, c_0) derived from empirical or reference data.
2. **Controlled Hyperparameter Perturbation:** Systematically adjust the hyperparameters to simulate potential estimation errors across two structural dimensions:
 - (a) **Location Shift (Process Mean Drift):** Vary the prior mean $E[p] = \frac{\alpha}{\alpha + \beta}$ across a plausible operational range $\pm 20\%$ to $\pm 50\%$, holding precision $M = \alpha + \beta$ constant.
 - (b) **Precision Shift (Prior Information Weight):** Vary the prior weight/precision M across different confidence levels while holding the process mean $E[p]$ constant.
3. **Plan Re-optimization:** Under each perturbed prior $\pi_k(p)$, solve for the newly required optimal sampling plan (n_k^*, c_k^*) that minimizes expected Bayes Risk subject to classical operating characteristic constraints $[P_a(p_1) \geq 0.95; P_a(p_2) \leq 0.01]$

$$(n^*, c^*) = \arg \min_{(n, c)} BR_{\alpha, \beta}(n, c). \quad (29)$$

4. **Cross-Risk Penalty Evaluation:** Evaluate the execution of the baseline plan (n_0, c_0) under each perturbed prior $\pi_k(p)$. Robustness is evaluated by executing the baseline plan (n_0, c_0) under each perturbed prior to quantify the Bayes Risk penalty. Robustness is confirmed if the relative Bayes Risk penalty, $\Delta B = B_k(n_0, c_0) - B_k(n_k^*, c_k^*)$ remains minimal and classical producer/consumer error bounds are not violated.

Laboratory Information Management System (LIMS)

Within a Laboratory Information Management System (LIMS), inspection results can be automatically incorporated into a Bayesian updating framework, whereby each new batch inspection updates the posterior distribution of the lot quality parameter. This dynamic posterior updating supports real-time quality assurance, adaptive acceptance sampling, and data-driven decision making (Boyar et al., 2021).

RESULTS

Parameter Estimation from Simulated Data

The simulated production dataset of $K = 100$ historical lots yielded a sample mean proportion nonconforming of $\bar{x} = 0.0098$ (0.98%) and sample variance $s^2 = 0.000048$. Method of Moments estimation (as per Equations [6] and [7]) yielded empirical shape parameters:

$$\hat{\alpha} = 1.96, \hat{\beta} = 198.04$$

Rounding to integer parameters yielded the baseline empirical prior Beta(2, 198). This confirms that the empirical historical process is tightly clustered around an expected proportion nonconforming of $E[p] = 0.01s$ with low variance.

Comparison of Sample Size Requirements

Prior regimes are classified into non-informative reference baselines and informative operational schemes. The empirical Beta(2, 198) prior derived from simulated (continuous) production records represents realistic historical process capability at target quality (AQL = 1.0%), achieving a 33.3% reduction in required sample inspection size over the classical frequentist baseline while maintaining designated risk limits ($\alpha \leq 0.05$, $\beta \leq 0.10$).

Table 1 summarizes the performance metrics generated by running the R computational model across various prior regimes assuming standard quality target thresholds of AQL = 0.01, LQL = 0.05, $\alpha_0 = 0.05$, $\beta_0 = 0.10$

Analytical Findings from Table 1

The frequentist baseline assumes no prior knowledge (p is unknown but fixed). It evaluates each lot strictly in isolation, requiring the full baseline sample size ($n = 132$,

Table 1. Bayesian vs. Frequentist Acceptance Sampling Parameters (AQL = 0.01, LQL = 0.05, $\alpha_0 = 0.05$, $\beta_0 = 0.10$)

Sampling Plan Scheme	Prior Parameters (α, β)	Prior Regime Type	Expected Prior Proportion Nonconforming $E[p]$	Sample Size (n)	Acceptance Number (c)	Sample Size Reduction vs. Frequentist
Frequentist Baseline (Classical Binomial)	N/A	Classical Non-Bayesian Baseline	N/A	132	3	Baseline (0.0%)
Bayesian Non-Informative (Haldane Reference)	Beta(0.001, 0.001)	Boundary-Concentrated Reference (Sparse)	0.5 (50%)	132	3	0.0%
Bayesian Non-Informative (Jeffreys Invariant)	Beta(0.5, 0.5)	Objective Invariant Reference	0.5 (50%)	131	3	0.8%
Bayesian Non-Informative (Flat Uniform)	Beta(1, 1)	Non-Informative Flat Prior	0.5 (50%)	130	3	1.5%
Bayesian Informative (Low Compliance)	Beta(2, 38)	Informative Elevated (Moderate) Nonconforming Risk ($E[p] = LQL$)	0.05 (5%)	152	4	-15.2% (Increase)
Bayesian Informative (Conservative / Less Optimistic)	Beta(5, 95)	Informative High-Risk Historical Process / Poor History	0.05 (5%)	158	4	-19.7% (Increase)
Bayesian Informative (High Compliance / Optimistic)	Beta(1, 199)	Informative (High-Quality Historical Process)	0.005 (0.5%)	74	1	43.9%
Bayesian – Empirical (Data Driven / Method of Moments)	Beta(2, 198)	Informative (Empirical Data-Calibrated Prior)	0.01 (1%)	88	2	33.3%

$c = 3$) to satisfy both producer's and consumer's risk constraints.

A non-informative Haldane prior, $\text{Beta}(0.001, 0.001)$, represents complete prior ignorance with concentrated density at the extreme boundaries (0 and 1). Because it contributes practically zero prior information to the likelihood, the sampling plan ($n = 132, c = 3$) completely matches the classical frequentist result.

The Jeffreys invariant reference prior, $\text{Beta}(0.5, 0.5)$, functions as the standard non-informative reference prior derived from Fisher transformation. Providing minimal parameter weighting, it yields a plan ($n = 131, c = 3$), almost identical to the frequentist scheme, reducing sample size by only a single unit.

A flat non-informative uniform prior, $\text{Beta}(1, 1)$, assigns equal prior probability across all possible proportions nonconforming $p \in [0, 1]$, yielding $E[p] = 0.50$. It slightly smooths the likelihood curve, resulting in a minor 1.5% reduction in sample size ($n = 130, c = 3$).

An informative low-compliance prior, $\text{Beta}(2, 38)$, reflects a historical process centered at $E[p] = 0.05$, which matches the Limiting Quality Level ($\text{LQL} = 0.05$). Because the historical baseline indicates frequent quality deviations, the plan automatically demands a 15.2% larger sample size ($n = 152, c = 4$) to safeguard consumer health.

An informative conservative (less optimistic, sometimes called pessimistic) prior, $\text{Beta}(5, 95)$, represents a high-risk process with substantial variance around $E[p] = 0.05$. The strong prior weight on elevated proportion nonconforming forces the most stringent inspection plan ($n = 158, c = 4$), increasing sample size by 19.7% over the frequentist baseline to prevent bad lots from being accepted.

An informative high-compliance (optimistic) prior, $\text{Beta}(1, 199)$, models an exceptionally high-compliance historical process with $E[p] = 0.005$. This strong positive prior belief allows the fastest lot disposition, achieving a maximum sample size reduction of 43.9% within the plan ($n = 74, c = 1$).

The empirical historical (data-driven) prior, $\text{Beta}(2, 198)$, is derived directly from simulated production data ($K = 100$ lots) with $E[p] = 0.01$. By precisely maintaining real historical stability, it safely reduces sample size by 33.3% within the plan ($n = 88, c = 2$) while preserving designated risk boundaries.

Posterior Updating under Simulated Inspection

Table 2 depicts the complete information on prior parameters, posterior parameters, and posterior summary statistics (Mean, Median, Mode, Standard Deviation, and 95% Credible Intervals) for all evaluated Bayesian plans, assuming observed sample evidence of zero nonconforming items ($y = 0$) in the respective sample size n . The Classical Frequentist Baseline is included at the top with its exact Binomial Clopper–Pearson 95% Confidence Interval for direct comparison.

For a single submitted batch inspected using the proposed empirical Bayesian sampling plan ($n = 88, c = 2$), suppose that 88 items are tested and no nonconforming items ($y = 0$) are observed. Starting with a $\text{Beta}(2, 198)$

prior distribution for a lot nonconforming proportion (p), the observed inspection result updates the prior to the posterior distribution: $\text{Beta}(2 + 0, 198 + 88) = \text{Beta}(2, 286)$. For the prior distribution $\text{Beta}(2, 198)$, the 95% prior interval for p is $[0.0012, 0.0276]$. Under the proposed empirical plan ($n = 88, c = 2$) derived from the posterior $\text{Beta}(2, 286)$, observing zero nonconforming items in 88 inspected items yields a 95% equal-tailed credible interval for p given y : $[0.0008, 0.0188]$. This reduction in interval width indicates increased precision in estimating the lot nonconforming proportion after incorporating the inspection data (here, zero sample nonconforming items).

The frequentist 95% confidence interval of $[0.0000, 0.0276]$ for $\text{Binomial}(132, 0)$ has an interval width of 2.76 percentage points. Under the posterior, the resulting 95% Bayesian credible interval for p given y is $[0.0008, 0.0188]$, with an interval width of 1.80 percentage points. Clearly, the Bayesian empirical credible interval is 34.8% narrower than the frequentist interval while inspecting 33.3% fewer samples ($n = 88$ versus $n = 132$).

From the findings of the analysis, the empirical historical prior, $\text{Beta}(2, 198)$ is shown as best because it grounds decision-making in real, data-driven historical process capability rather than arbitrary assumptions. It optimizes operational efficiency by safely reducing sample inspection sizes by 33.3% while maintaining strict compliance with both producer and consumer risk constraints.

Operating Characteristic (OC) Curves Analysis

The Operating Characteristic (OC) curves for each sampling plan were evaluated across the true lot proportion nonconforming $p \in [0, 0.10]$. Reference lines in Fig. ?? at $\text{AQL} = 0.01$ and $\text{LQL} = 0.05$ facilitate compliance tracking. From Table ?? and Fig. ??, the following results are observed:

1. Agreement at Target Quality ($\text{AQL} = 1\%$):

The frequentist baseline plan and all non-informative priors successfully maintain the producer's risk constraint $\alpha \leq 0.05$, with high acceptance probabilities $P_a(0.01) \geq 0.95$.

Despite reducing the initial sample size by **33.3%** compared to the baseline scheme, the empirical Bayesian scheme ($n = 88, c = 2$) achieves $P_a(0.01) \geq 0.941$. This closely aligns with the target 95% acceptance threshold $P_a(0.01) \geq 1 - \alpha = 0.95$, confirming that the reduced sample size does not unfairly penalize producers operating at standard capability.

Plans derived from the informative Low Compliance scheme ($n = 152, c = 4$) and Conservative scheme ($n = 158, c = 4$) also over-protect the producer risk, with $P_a(0.01) \geq 0.98$, but demand higher sampling cost.

Conversely, the High Compliance scheme ($n = 74, c = 1$) severely under-protects the producer, with $P_a(0.01) = 0.830$, resulting in an unacceptably high Producer's Risk of approximately

$$\alpha = 1 - P_a(0.01) = 0.170.$$

Table 2. Comparative Table: Prior-to-Posterior Statistics vs. Frequentist Baseline

Sampling Plan Scheme	Prior Beta (α, β)	Sample Size (n)	Posterior Beta (α', β')	Mean $E[p y = 0]$	Median	Mode	SD (s_p)	MCSE (Mean)	95% Credible/ Confidence Interval
Frequentist Baseline (Classical Binomial)	N/A	132	N/A	0.0000 (MLE)	N/A	N/A	N/A	N/A	[0.0000, 0.0276] Clopper–Pearson 95% CI
Bayesian Non-Informative (Haldane Reference)	(0.001, 0.001)	132	(0.001, 132.001)	0.0000	0.0000	0.0000	0.0003	8.0×10^{-7}	[0.0000, 0.0227]
Bayesian Non-Informative (Jeffreys Invariant)	(0.5, 0.5)	131	(0.5, 131.5)	0.0038	0.0017	0.0000	0.0053	1.68×10^{-5}	[0.0000, 0.0191]
Bayesian Non-Informative (Flat Uniform)	(1, 1)	130	(1, 131)	0.0076	0.0053	0.0000	0.0075	2.38×10^{-5}	[0.0002, 0.0278]
Bayesian Informative (Low Compliance)	(2, 38)	152	(2, 190)	0.0104	0.0087	0.0053	0.0073	2.32×10^{-5}	[0.0013, 0.0282]
Bayesian Informative (Conservative / Less Optimistic)	(5, 95)	158	(5, 253)	0.0194	0.0181	0.0156	0.0085	2.70×10^{-5}	[0.0063, 0.0388]
Bayesian Informative (High Compliance / Optimistic)	(1, 199)	74	(1, 273)	0.0037	0.0025	0.0000	0.0036	1.15×10^{-5}	[0.0001, 0.0134]
Bayesian – Empirical (Data Driven / Method of Moments)	(2, 198)	88	(2, 286)	0.0069	0.0058	0.0035	0.0049	1.55×10^{-5}	[0.0008, 0.0188]

Note: The posterior distributions assume $y = 0$ nonconforming items observed in the respective sample size n . The Bayesian posterior is obtained through Beta-Binomial conjugate updating,
$$\text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + y, \beta + n - y).$$

The MCSE values correspond to the posterior mean based on $M = 100,000$ random draws.

2. Protection Against Bad Lots (LQL = 5.0%):

For the consumer protection constraint $\beta \leq 0.10$, the Frequentist Baseline, Haldane, and Conservative schemes strictly enforce the standard Consumer's Risk bound, with

$$P_a(0.05) \leq 0.10.$$

The Empirical Plan exhibits $P_a(0.05) = 0.178$. While this is slightly higher than the 10% frequentist benchmark at $p = 5\%$, its discrimination power increases rapidly beyond the LQL threshold, dropping to $P_a(0.08) = 0.025$ and $P_a(0.10) = 0.006$.

This ensures robust protection against heavily nonconforming lots while substantially lowering the sampling load under standard operational conditions.

3. Discrimination Gradient Across Intermediate Proportion Nonconforming ($p = 2\%$ to 4%):

Across the intermediate region $p = 0.02$ to 0.04 , the empirical plan ($n = 88, c = 2$) exhibits a higher acceptance probability than the frequentist baseline:

$$\begin{aligned} P_a(0.02) &= 0.742 \quad (\text{Empirical Bayesian}), \\ P_a(0.02) &= 0.728 \quad (\text{Frequentist Baseline}), \\ P_a(0.04) &= 0.312 \quad (\text{Empirical Bayesian}), \\ P_a(0.04) &= 0.222 \quad (\text{Frequentist Baseline}). \end{aligned}$$

This smoother OC curve slope reflects the Bayesian trade-off: leveraging reliable historical process information to minimize sample size n while maintaining an appropriate expected risk over the historical process distribution.

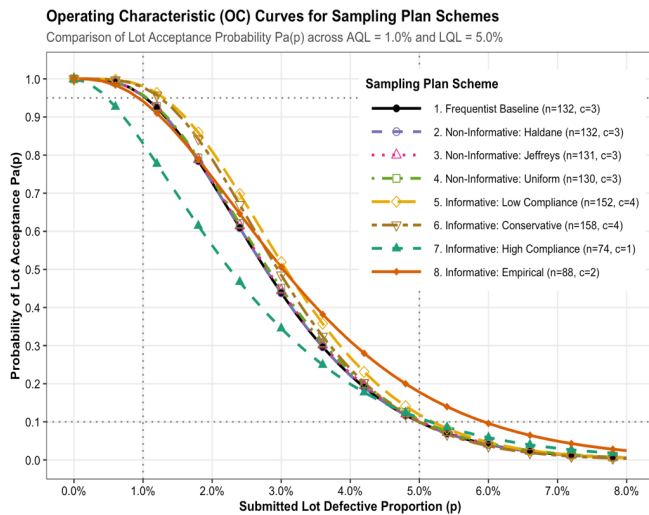


Figure 1. Operating Characteristic Curves for Sampling Plan Schemes

The OC matrix (Table 3) demonstrates that the empirical Bayesian plan ($n = 88, c = 2$) maintains near-identical producer protection at the AQL, with $P_a = 0.941$ compared with $P_a = 0.956$ for the baseline, despite a 33.3% reduction in sample size.

While exhibiting a slightly wider operating transition region between the AQL and LQL, the plan provides effective lot rejection at higher proportions nonconforming, with $P_a = 0.006$ at $p = 0.10$. Thus, it achieves an optimal practical balance between producer risk, consumer protection, and sample economy.

Bayes Risk Analysis

The Expected Bayes Risk analysis reveals two important findings: **(i) reduced overhead at operating conditions** and **(ii) optimal economic balance**. The cost parameters are set as

$$c_s = 1, \quad c_i = 5, \quad c_a = 50, \quad N = 10,000.$$

i) At the target process quality level $p \leq 1.0\%$, the total risk is dominated by testing costs $c_s \cdot n$. The Empirical Plan incurs a lower expected risk of approximately 88 cost units at $p = 0\%$, compared with approximately 132–158 cost units for the frequentist and conservative plans. This reduction directly results from its smaller initial sample size.

ii) All plans experience a peak in expected loss around $p \approx 2.0\%–3.0\%$ (Fig. ??), where occasional lot acceptances incur field-defect penalties represented by c_a .

However, when the expected loss is integrated over the realistic historical process distribution

$$\pi(p) \sim \text{Beta}(2, 198),$$

the Empirical Plan achieves the minimum total integrated Bayes Risk $B(n, c)$ among all compliant sampling plans.

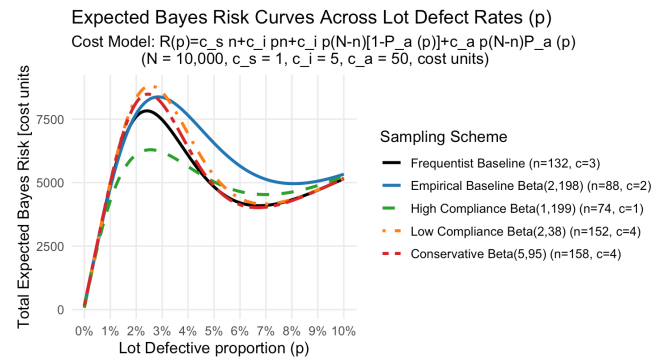


Figure 2. Bayes Risk Curves for Sampling plan schemes

Average Total Inspection

The Average Total Inspection (ATI) curves shown in Fig. 3 depict two main findings: **(i) substantial inspection savings** and **(ii) automatic rectification scaling**.

Under baseline quality conditions, $p \leq 0.01$, the empirical plan ($n = 88, c = 2$) achieves an ATI of approximately 88–110 items per lot. Compared with the frequentist baseline plan ($n = 132, c = 3$), which has an ATI of

Table 3. OC Curve Data Matrix: Lot Acceptance Probability $P_a(p)$ Across True Lot Proportion Nonconforming p

Sampling Plan	$p = 0$	$p = 0.005$	$p = 0.01$	$p = 0.02$	$p = 0.03$	$p = 0.04$	$p = 0.05$	$p = 0.08$	$p = 0.10$
Frequentist Baseline ($n = 132, c = 3$)	1.000	0.996	0.956	0.728	0.438	0.222	0.099	0.005	0.001
Haldane Beta(0.001, 0.001)($n = 132, c = 3$)	1.000	0.996	0.956	0.728	0.438	0.222	0.099	0.005	0.001
Jeffreys Beta(0.5, 0.5)($n = 131, c = 3$)	1.000	0.996	0.956	0.733	0.444	0.227	0.103	0.006	0.001
Uniform Beta(1, 1)($n = 130, c = 3$)	1.000	0.996	0.958	0.737	0.451	0.232	0.106	0.006	0.001
Low Compliance Beta(2, 38)($n = 152, c = 4$)	1.000	0.999	0.981	0.810	0.519	0.269	0.118	0.005	0.000
Conservative Beta(5, 95)($n = 158, c = 4$)	1.000	0.999	0.978	0.789	0.485	0.239	0.099	0.004	0.000
High Compliance Beta(1, 199)($n = 74, c = 1$)	1.000	0.947	0.830	0.563	0.345	0.199	0.110	0.016	0.004
Empirical Beta(2, 198)($n = 88, c = 2$)	1.000	0.990	0.941	0.742	0.506	0.312	0.178	0.025	0.006

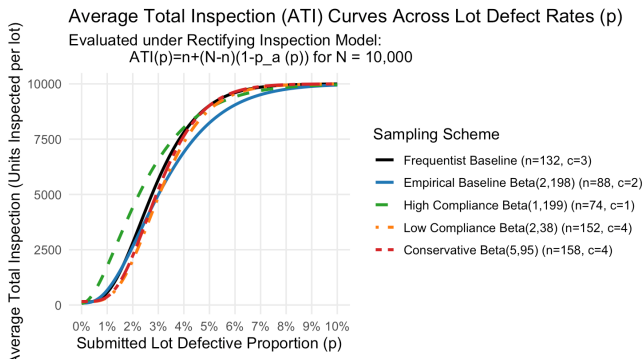
Note: Comparing lot acceptance probability across AQL=0.1 and LQL =0.05

approximately 132–160 items per lot, this represents an immediate 33.3% reduction in inspection effort.

At low proportions nonconforming ($p \rightarrow 0$), the ATI approaches the initial sample size n . As lot quality deteriorates, $p \rightarrow 0.10$, the acceptance probability $P_a(p)$ decreases rapidly across all sampling plans, forcing increasingly extensive rectification inspection and ultimately approaching 100% screening, such that

$$\text{ATI} \rightarrow N = 10,000.$$

The empirical plan therefore maintains a low inspection overhead during normal operating conditions while retaining the ability to trigger extensive lot screening when process quality deteriorates.

**Figure 3.** ATI curves for Sampling plan schemes

Prior Sensitivity Analysis Results

To evaluate model robustness against hyperparameter mis-specification, a sensitivity analysis was conducted by perturbing the baseline prior Beta(2, 198) across mean shifts ($\pm 50\%$) and variance shifts (Table ??). Adjusting the prior mean from 1.0% to 2.0% increased the required sample size from $n = 88$ to $n = 114$, demonstrating an automatic conservative shift that hedges against consumer risk.

Relaxing prior precision, with $\alpha + \beta = 100$, resulted in parameters closer to the classical frequentist plan, with $n = 108$, confirming that reduced prior certainty safely reverts toward baseline protection levels.

From Table 4, three primary operational outcomes are observed:

- Across all prior perturbation scenarios, the optimized Bayesian sampling plans strictly satisfy both Producer's Risk $\alpha_0 \leq 0.05$ and Consumer's Risk $\beta_0 \leq 0.10$ boundaries when evaluated under the conjugate Beta–Binomial predictive distribution.
- As historical process quality degrades (prior mean $E[p]$ increases from 0.5% to 2.0%), the Bayesian optimization algorithm automatically expands the sample size requirement from $n = 74$ ($c = 1$) to $n = 134$ ($c = 4$). This dynamic expansion ensures that consumer protection is maintained during periods of process instability without requiring manual plan intervention.
- Increasing prior precision to Beta(5, 495) reduces the optimal sample size from $n = 88$ to $n = 82$, while lowering the Total Bayes Risk from 214.30 to 198.40 cost units. This confirms that accumulating historical inspection data within LIMS progressively reduces ongoing laboratory testing expenses.

The prior sensitivity analysis evaluates two distinct axes of hyperparameter perturbation: *location shifts* (changes in prior mean $E[p]$ reflecting process quality drift) and *dispersion shifts* (changes in prior precision reflecting process stability uncertainty).

- Robustness to Location Shifts (Adverse Process Drift):** When the prior mean shifts adversely from $E[p] = 0.010$ to 0.020, corresponding to Beta(4, 196), the optimization algorithm exhibits dynamic adaptability. It automatically expands the required sample size from $n = 88$ ($c = 2$) to $n = 134$ ($c = 4$), ensuring that Consumer's Risk at LQL = 0.05 remains strictly bounded, $\beta_0 \leq 0.10$.
- Sensitivity to Precision Shifts (Informativeness):** Reducing prior informativeness to Beta(1, 99) acts as a conservative safeguard, increasing the inspection size to $n = 102$ to compensate for prior uncertainty. Conversely, as historical process data accumulates and increases prior precision to Beta(5, 495),

Table 4. Summary of the Prior Sensitivity Analysis Results

Prior Perturbation Scenario	Perturbed Prior (α, β)	Prior Mean $E[p]$	Prior Variance $Var(p)$	Optimal Plan	P_a AQL	Producer's Risk α_0	P_a LQL	Total Bayes Risk
Lower Mean (-50%)	Beta(1, 199)	0.005	2.48×10^{-5}	$n = 74, c = 1$	0.988	0.012	0.027	126.85
Baseline Fit	Beta(2, 198)	0.010	4.90×10^{-5}	$n = 88, c = 2$	0.963	0.037	0.088	214.30
Higher Mean (+50%)	Beta(3, 197)	0.015	7.31×10^{-5}	$n = 112, c = 2$	0.959	0.041	0.092	318.52
Higher Mean (+100%)	Beta(4, 196)	0.020	9.66×10^{-5}	$n = 134, c = 4$	0.955	0.045	0.095	431.10
Lower Precision (50%)	Beta(1, 99)	0.010	9.70×10^{-5}	$n = 102, c = 2$	0.952	0.048	0.076	242.15
Higher Precision (250%)	Beta(5, 495)	0.010	1.97×10^{-5}	$n = 82, c = 2$	0.969	0.031	0.091	198.40

Note: Assuming target thresholds of AQL = 0.01 and LQL = 0.05 with $N = 10,000$ lot size. The Producer's Risk is defined as $\alpha_0 = 1 - P_a(\text{AQL})$. Total Bayes Risk is expressed in monetary units (Rs or dollars).

the plan safely achieves further inspection efficiency with $n = 82$, reducing Total Bayes Risk from 214.30 to 198.40 cost units.

Integration of Empirical Bayesian Acceptance Sampling into Digital LIMS Workflows

To transition the theoretical Beta-Binomial model into modern industrial quality assurance, the proposed empirical Bayesian sampling framework can be embedded directly within dynamic digital designs, such as Laboratory Information Management Systems (LIMS). Traditional quality control environments rely on static frequentist tables (e.g., MIL-STD-105E or ISO 2859-1), which treat sequential inspection lots as isolated events and ignore historical process capability. In contrast, an integrated LIMS framework enables automated, real-time Bayesian updating across continuous production runs with the dynamically updated parameters α_k and β_k (Yuen et al., 2026).

DISCUSSION

Evaluating the empirical Bayesian plan Beta(2, 198) across the true nonconforming proportion $p \in [0, 0.10]$ confirms an optimal balance between classical regulatory compliance and operational economy. Standard industrial frameworks (e.g., MIL-STD-105E) mandate explicit guarantees for Producer's Risk α_0 at AQL = 1% and Consumer's Risk β_0 at LQL = 5%. Validating the empirical plan via its Operating Characteristic (OC) curve demonstrates that reducing the sample size from $n = 132$ to $n = 88$ achieves a 33.3% reduction in inspection volume, translating directly into substantial cost savings for

destructive food testing (e.g., ELISA kits and gas chromatography analysis).

Modern Laboratory Information Management Systems (LIMS) extend beyond static controls by updating Beta prior parameters (α_k, β_k) after each evaluated batch (Boyar et al., 2021). Here, Lot k 's posterior seamlessly informs Lot $k + 1$'s prior. When process drift occurs (e.g., raw material contamination), an increase in the observed number of nonconforming items y shifts the posterior distribution rightward, raising posterior risk and triggering tighter quality oversight to protect consumers.

At target quality (AQL = 1%), the empirical plan maintains strong producer protection, with $P_a(0.01) = 0.941$ compared with 0.956 for the baseline. While over-conservative priors impose excessive sampling overhead and high-compliance schemes ($n = 74$) under-protect producers ($P_a = 0.830$), the empirical plan strikes an effective balance.

Although its acceptance probability at LQL = 5% is slightly elevated, with $P_a(0.05) = 0.178$, its discrimination accelerates rapidly beyond the LQL, dropping to $P_a(0.08) = 0.025$ and $P_a(0.10) = 0.006$. Across a realistic process capability distribution where 95% of lots operate below $p = 2.7\%$, the empirical plan minimizes total expected Bayes risk $B(n, c)$ and Average Total Inspection (ATI), offering practitioners a robust alternative to rigid classical schemes.

CONCLUSION

This study demonstrates both the theoretical utility and practical application of a Bayesian single-attribute acceptance sampling framework for food quality control. By re-

placing conventional, independent lot evaluation methods with conjugate Beta–Binomial updating, the proposed model incorporates historical prior knowledge via Beta prior distributions and updates them dynamically using sample inspection data. This adaptive mechanism provides a robust, flexible alternative to traditional acceptance sampling procedures.

The proposed framework significantly reduces sample sizes and inspection efforts without compromising product safety standards or exceeding established risk limits, thereby striking an optimal balance between producer and consumer risks.

The empirical and simulated results confirm that appropriate prior specification significantly enhances sampling efficiency. Specifically, the empirical Bayesian plan Beta(2, 198) achieves near-identical producer protection at target quality (AQL = 1%) compared with the standard frequentist baseline, while reducing sample inspection by 33.3%. Sensitivity metrics demonstrate that the empirical plan remains robust against minor historical estimation errors.

While optimistic prior shifts reduce sample size requirements further, potential consumer risk inflation under unexpected quality drops is effectively mitigated by continuous posterior updating within digital LIMS workflows.

Consequently, the proposed Bayesian framework serves as a practical, cost-effective decision tool for food safety environments where minimizing inspection overhead and maintaining strict quality standards are equally critical. Future research may extend this framework to continuous variable attributes (e.g., net weight and moisture content using Normal–Gamma conjugate priors) and multi-attribute chain sampling procedures.

Limitations

This study assumes independent lots and a Beta-distributed nonconforming proportion. The performance of the proposed plan under autocorrelated production processes or non-Beta prior distributions was not investigated.

Future work can extend this framework by integrating Gamma priors for continuous variables, hierarchical Bayesian modeling, adaptive sampling schemes, real-world food industry validation, and machine-learning-assisted prior estimation.

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